Fundamentals of Networking

Final Project

Francisco Acosta | NETW191 | Feb 2024

Introduction

The fundamentals of networking module cover topics such as IP addresses, subnetting, connectivity testing, loopback interfaces, Visio network diagrams, and wireless network security.

This project showcases my knowledge of network management, building of key infrastructure components, troubleshooting device connectivity within a virtual environment, planning network architecture, and various network security tactics.

Follow along my journey navigating network configurations utilizing the command line interface (CLI), router settings and networking tables.



IPv4 Addressing

This section of the project was geared towards network support with a focus on the Implementation of an IPv4 addressing scheme.

The topic of network support was introduced through the practical application of assigning an IPv4 address to a small office/home office (SOHO) router in a virtual environment utilizing the Hyper-V Manager.

This endeavor highlighted the importance of IP addresses and the skills it takes to manage all associated tasks.

IPv4 Addressing

Changing a SOHO router's IPv4 address



The goal was to utilize the Hyper-V Manager to access and change a small office/home office (SOHO) router's IPv4 address via a virtual machine.



The initial step was to find the current gateway's IPv4 address utilizing a virtual machine's terminal.



The next step was to log into the SOHO’s router to change the root password.

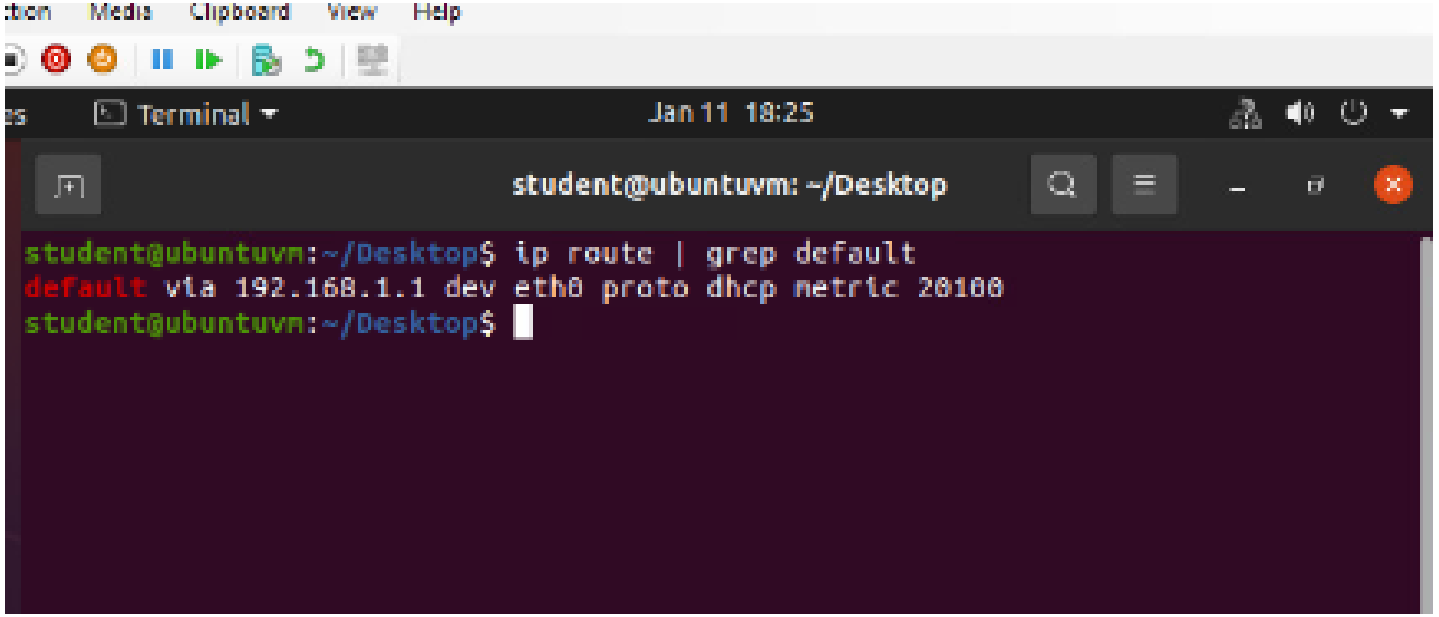


This was followed up by accessing the SOHO router with its new password on a virtual machine to change the IPv4 address and netmask.



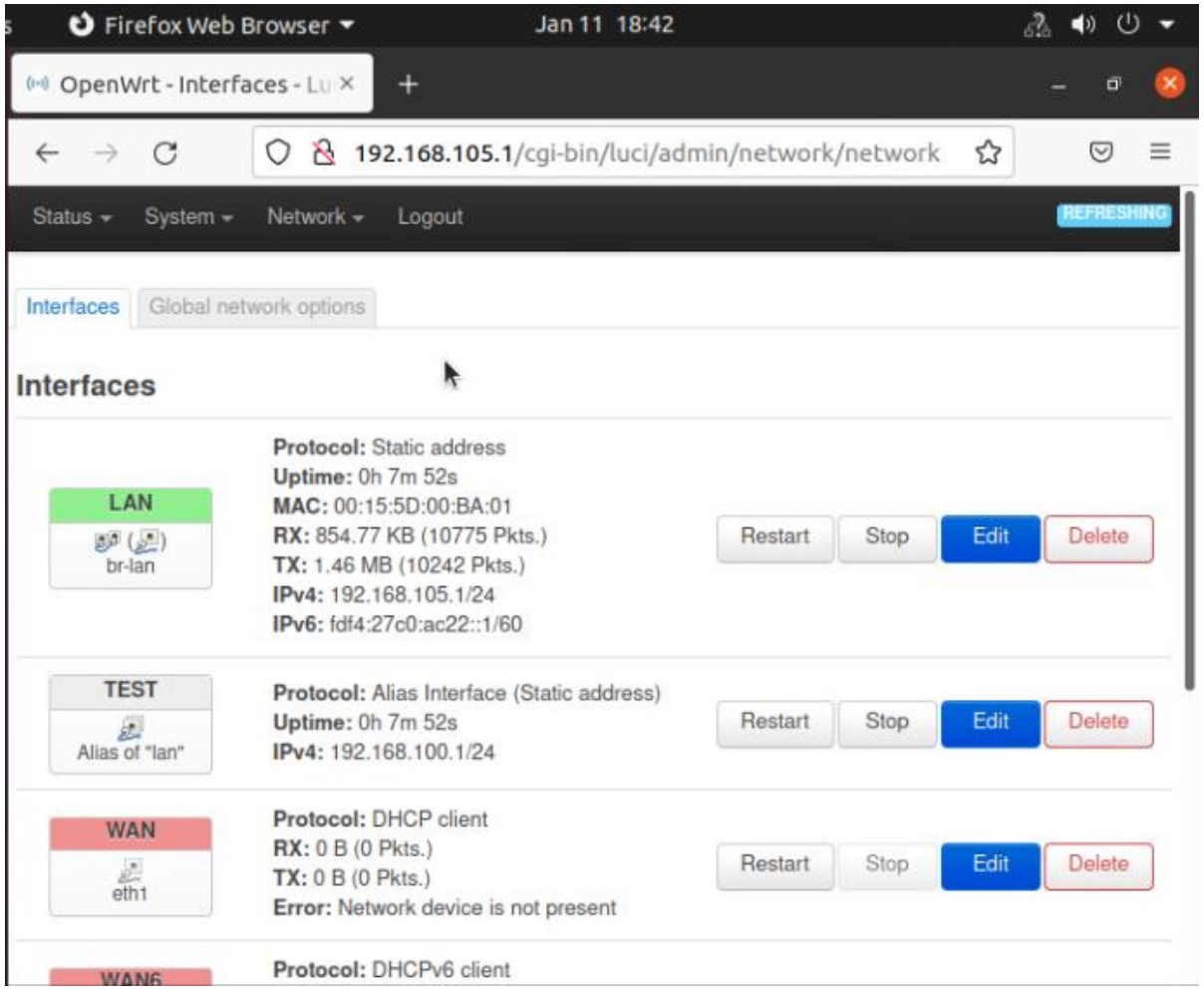
The final step was to utilize a browser to access and confirm the changes to the SOHO's router.

Preparation



The “ip route | grep default” command was utilized to find the default IP address of the SOHO Router LAN interface.

IPv4 Address Assignment



This section assigned a new IPv4 address and netmask to the SOHO router.



Connectivity Testing

This section focused on connectivity testing between two computers and a SOHO router in a virtual environment.

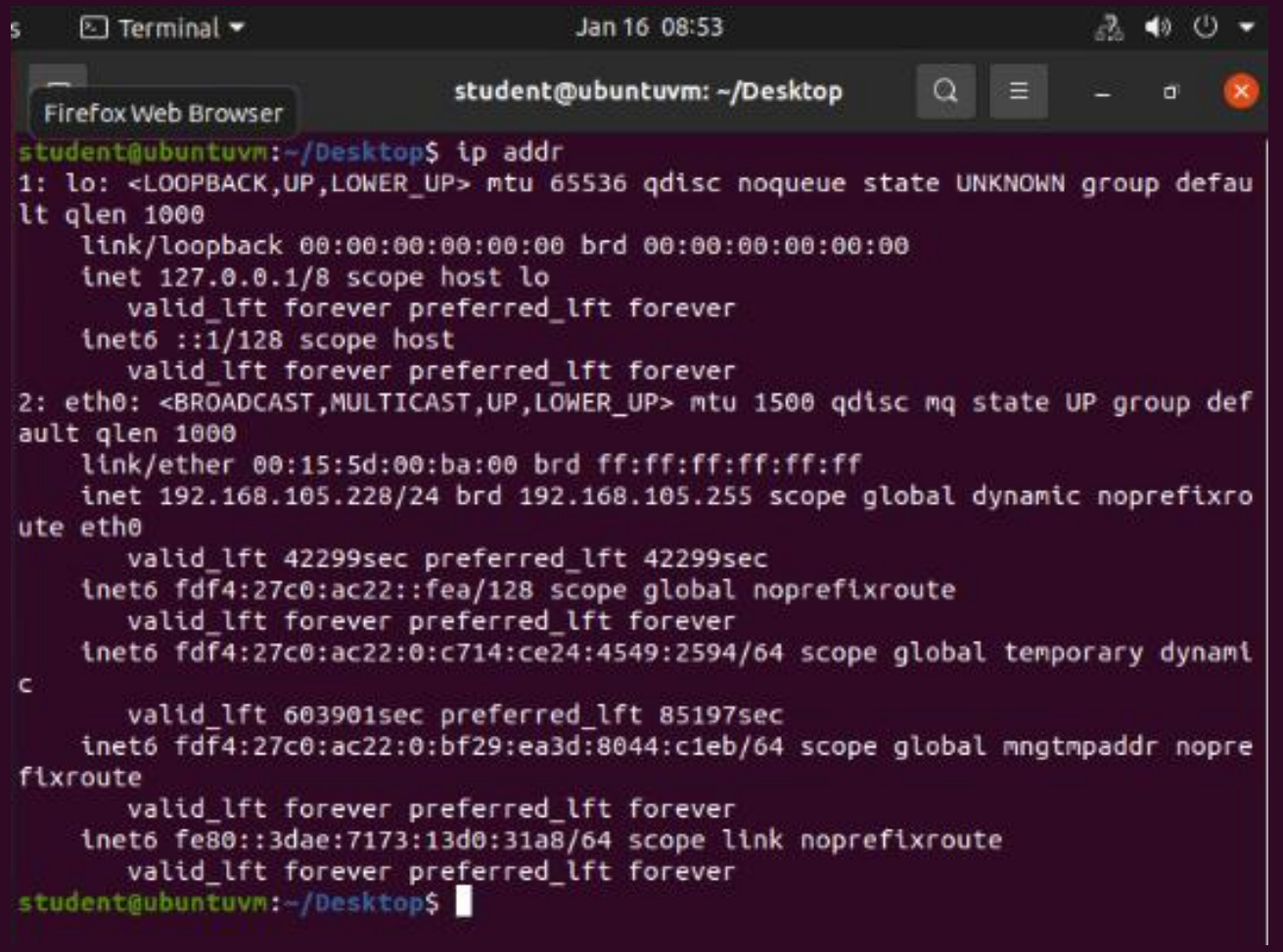
The process started by the creation of two virtual machines and a virtual SOHO router with the Hyper-V Manager.

The next phase included finding both virtual machine’s IP addresses with the “ip addr” command within the terminal of each machine.

The last phase included running the “ping” command on both virtual machines in their individual terminals to check the connectivity to each computer and the connectivity to the SOHO router.

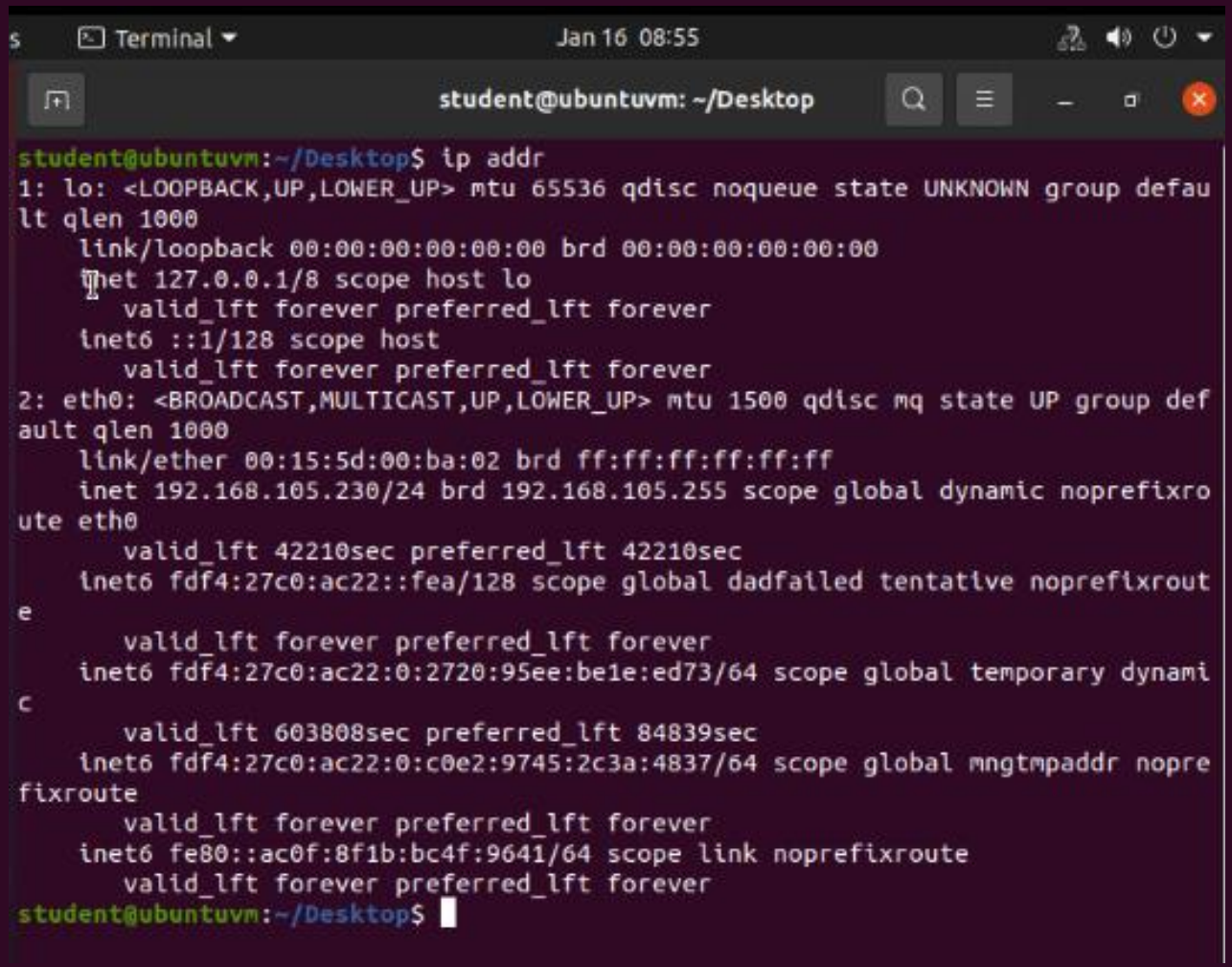
Dynamic IP Address Assignment

IPv4 address of the first virtual computer



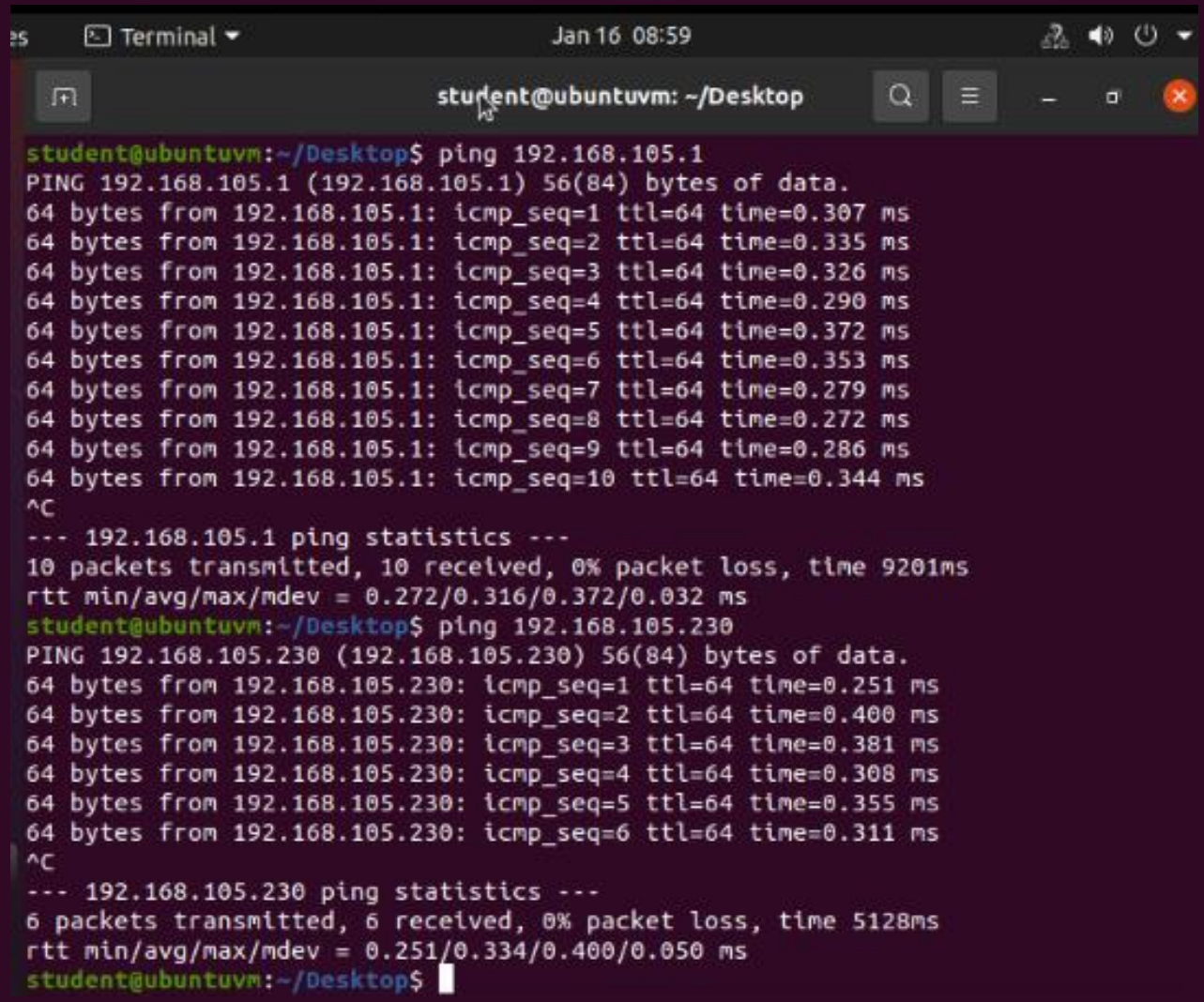
Dynamic IP Address Assignment

IPv4 address of the second virtual computer



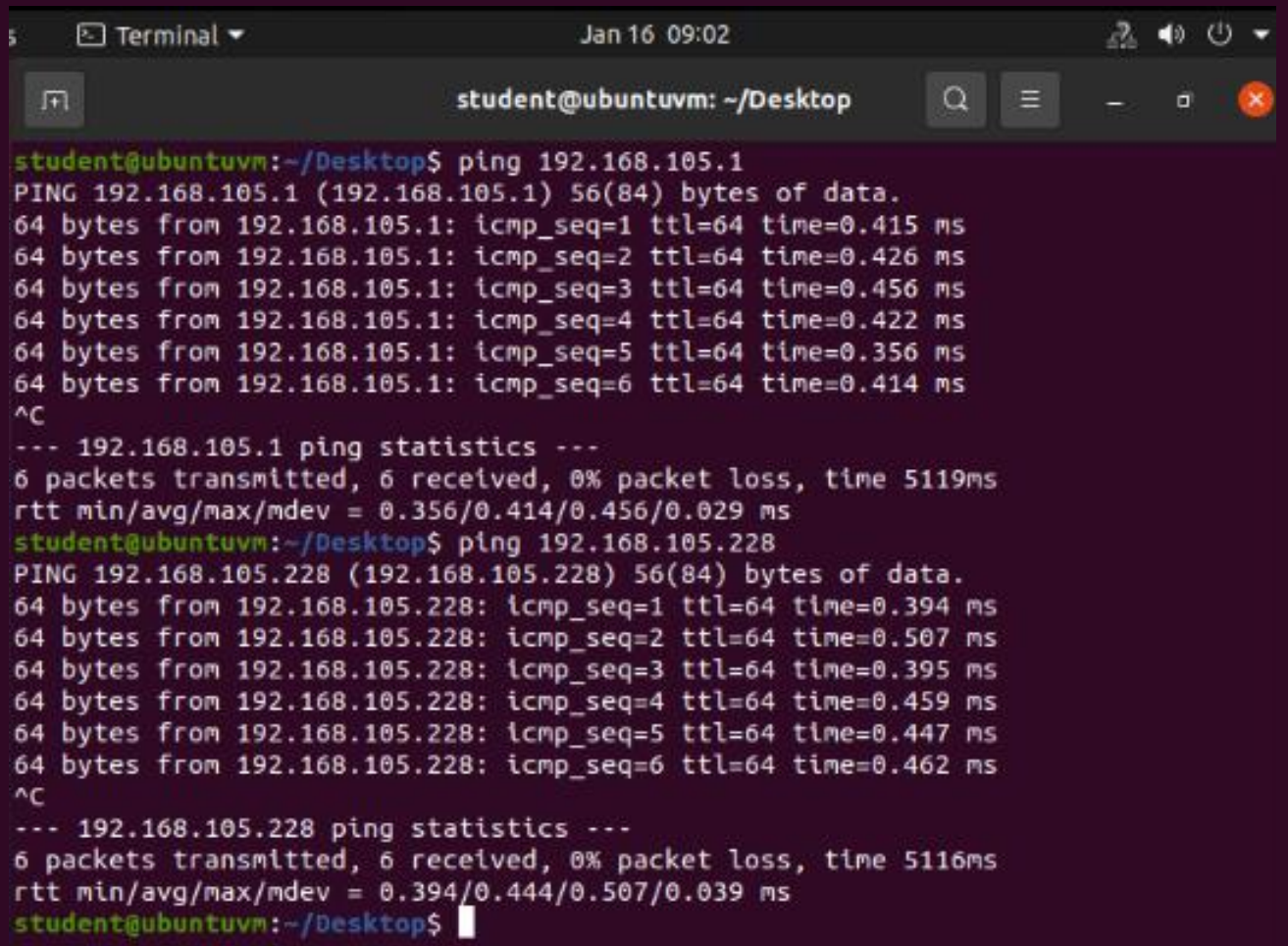
Connectivity Testing

Connectivity testing from the first virtual computer to the second virtual computer and virtual SOHO router



Connectivity Testing

Connectivity testing from the second virtual computer to the first virtual computer and virtual SOHO router





IP Subnetting and Loopback Interfaces

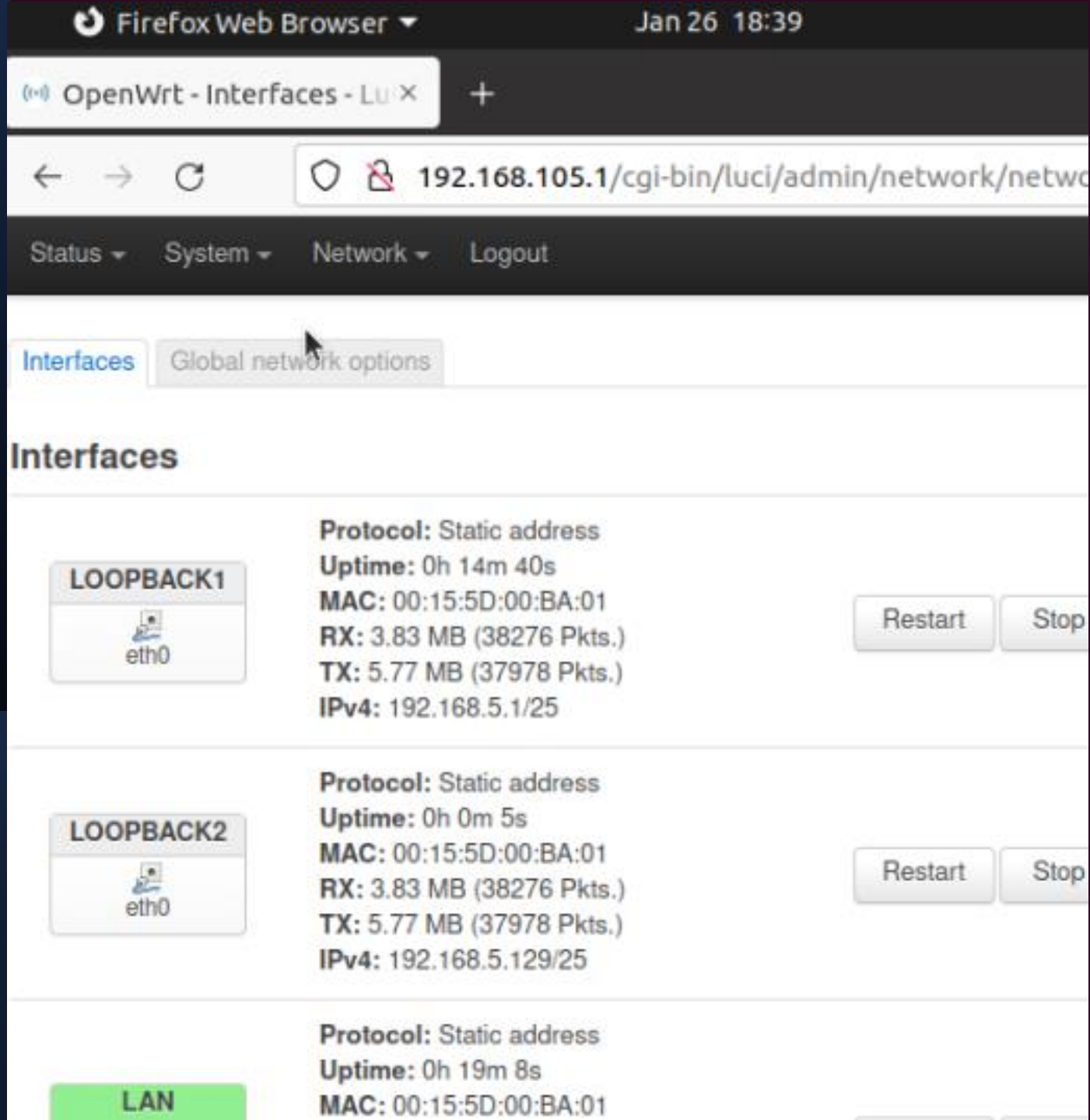
This section unraveled the world of subnetting and how to expand a network architecture through segmentation.

Steps included calculating subnet masks, identifying IP addresses, and exploring which host addresses were usable. Calculations were utilized to create two loopbacks via a web browser and were configured into usable network addresses that later underwent connectivity testing. Connectivity testing utilized the “ping” command via the terminal to verify those hosts were indeed usable.

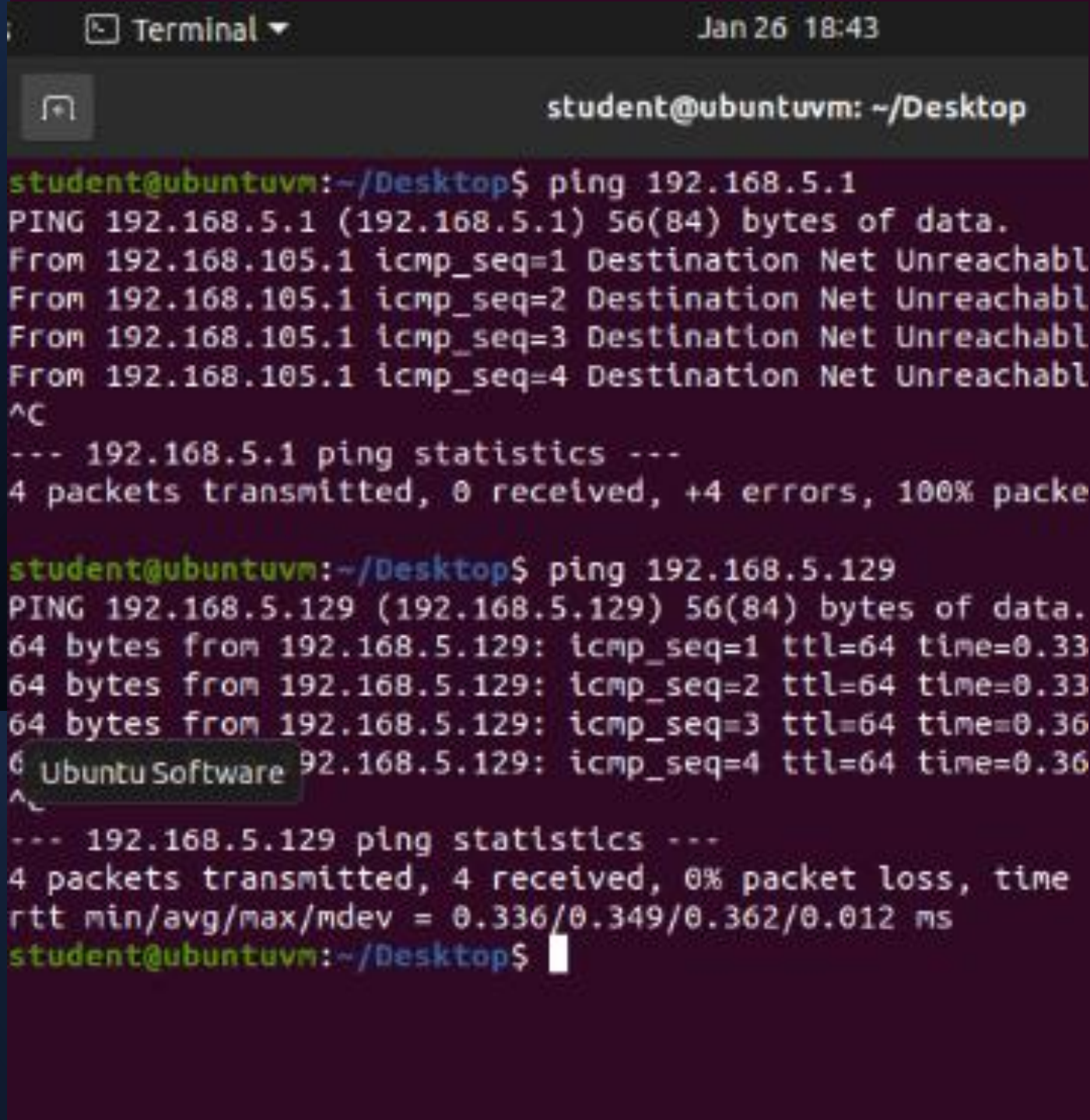
IP Subnet Calculations

Subnet ID	Network Mask (/prefix)	Network Mask (Dotted decimal)	Network Address	First Usable Host Address	Last Usable Host Address	Broadcast Address
0	/25	**255.255.255.128**	192.168.5.0	192.168.5.1	**192.168.5.126**	192.168.5.127
1	**/25**	255.255.255.128	**192.168.5.128**	192.168.5.129	192.168.5.254	**192.168.5.255**

Loopback Interfaces



Connectivity Testing





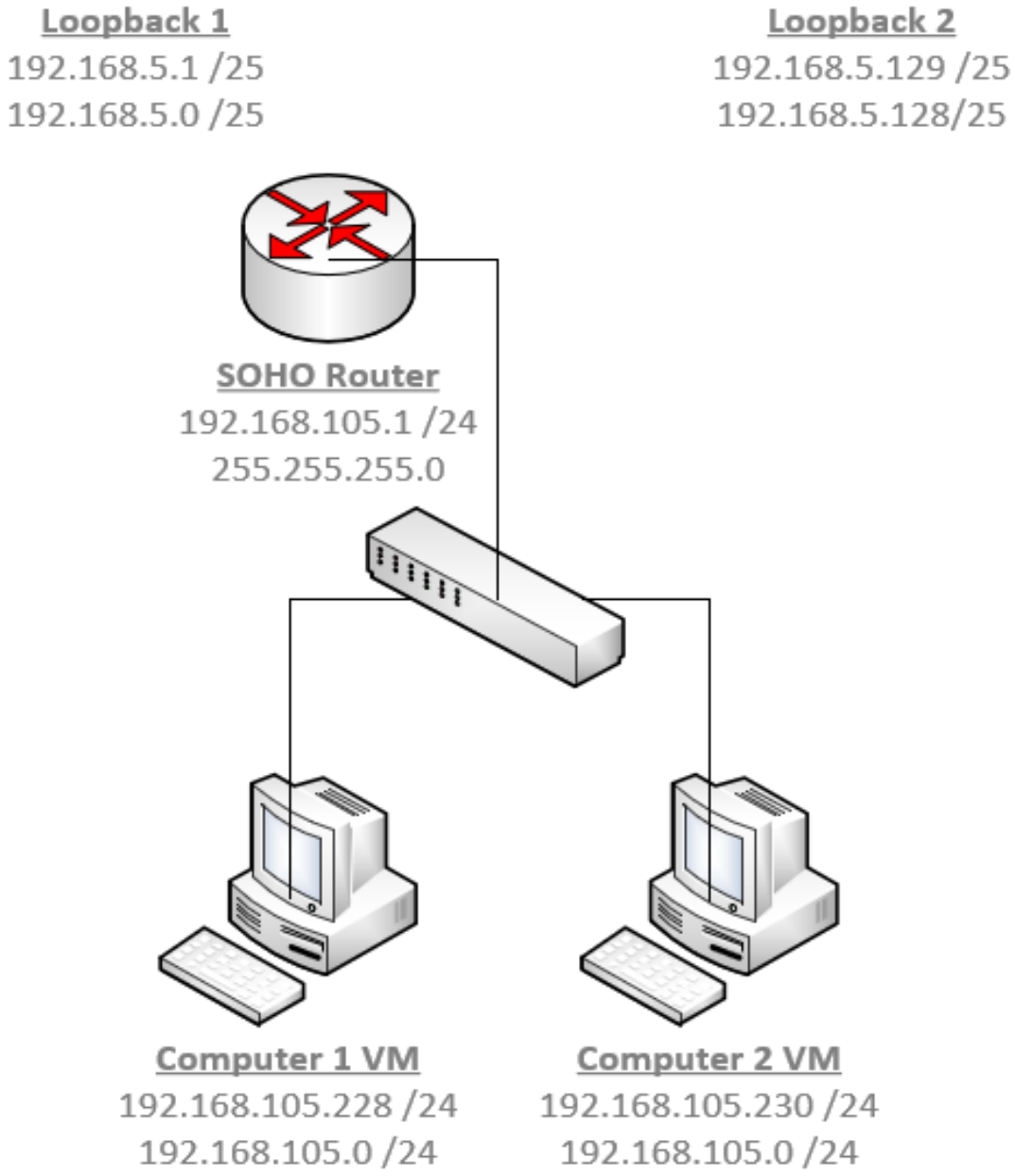
Visio Network Diagram

This section was centered around the creation of a Microsoft Visio network diagram. The creation of this diagram gave me the ability to visualize the network structure by keeping track of IP addresses, subnet masks, connections, and device types.

The goal was to illustrate the network with the correct device connections and associated naming schemes with the proper icons. It is a great way to organize and quickly understand how the network operates.

The network consists of two virtual machines, a virtual switch, a SOHO router, and two loopbacks.

Microsoft Visio Network Diagram



FA



SOHO Wireless Network Security

Local network management techniques to secure a wireless network was the focal point of this section.

I was shown how to properly secure a SOHO wireless network by changing default credentials such as usernames/password of a SOHO router, properly implement static or dynamic IP addresses, implement MAC filtering, and adjust settings to properly authenticate a wireless network connection.

SOHO Wireless Network Security 1 of 3

1. What is the factory default username and password of a TP-Link router? Why is it important to change the default username and password of a SOHO router?

- **The factory default username and password for a TP-Link router are both “admin”.**
- **It is important to change the default username and password of a SOHO router to eliminate security threats. These default credentials are listed on the manufacture’s website of a router and criminals can access your network with this commonly known knowledge.**

2. To protect a SOHO wireless network with a small number of devices, which address management method provides more control, configuring the device IP addresses manually (static IP) or using a DHCP server (dynamic IP)? Why?

- **A static IP address provides more control because a network administrator can manually assign specific IP addresses to devices on a network. This differs from a DHCP server which automatically assigns different addresses each time a device joins the network, which ultimately takes control away from the network administrator. The network administrator may also track and better control a network by knowing what each device is with its definitive static IP address, compared to a DHCP server that houses a pool of IP addresses.**

SOHO Wireless Network Security 2 of 3

3. What does MAC filtering do? If needed, when would you use deny filtering rules and when would you use allow filtering rules? What happens to devices that want to connect, if the "Allow the stations specified by any enabled entries in the list to access" function is enabled but there are no entries in the list?

- **MAC filtering allows a network administrator to control a wireless device's ability to gain access to a network by only allowing specified IP addresses through the the access point.**
- **I would use deny filtering rules if I wanted to block specific devices from a network.**
- **I would use allow filtering rules if I had a specified list of devices that I wanted to allow access to a network.**
- **No device would be allowed to join the network if the "Allow the stations specified by any enabled entries in the list to access" function is enabled with no entries. Adding entries would then grant those specific addresses access to the network.**

4. What wireless security settings are displayed on the Wireless Security page? Which one is recommended by the vendor? Why?

- **Disable Wireless Security, WPA/WPA2 – Personal, WPA/WPA2 – Enterprise, and WEP settings are displayed on the Wireless Security page.**
- **The vendor recommends "WPA/WPA2 – Personal" because it encrypts data by using a pre-shared key.**

SOHO Wireless Network Security 3 of 3

5. Among the configurations you explored in this module, which one is a true security function? Why?

- **Updating a wireless network's security settings to utilize "WPA/WPA2 – Personal" is a true security function because it encrypts data within a network.**

6. What would you do to protect your wireless network at home? Why?

- **I would change my router's username and password to differentiate it from the manufacture's default values which will increase the security of my network. I am also considering utilizing MAC filtering since the number of devices in my home network is minimal and does not change.**

Challenges

Challenges throughout this project included:

- familiarizing myself with locations to access various options such as changing an IPv4 netmask
- understanding the roles of virtual environment devices
- implementation of CLI commands to complete vital tasks
- gaining an understanding of the various parts of a network and the associated interconnectivity
- combining the basics of network security and putting it to practical use when navigating the SOHO router settings
- learning to calculate IP addresses and network masks

Career Skills

Skills acquired during this module include

- subnetting
- updating a network's architecture
- managing a network in an organized and efficient manner
- Utilizing Microsoft Visio
- managing IPv4 addresses
- navigating a virtual environment with the Hyper-V Manager
- implementation of command line scripts
- testing the connectivity between devices
- establishing the correct authentication settings
- applying MAC filtering

Conclusion

This project covered the fundamentals of networking with topics such as IPv4 addressing, connectivity testing, IP subnetting, loopback interfaces, creation of Microsoft Visio diagrams, and SOHO wireless network security.

Networking is a crucial part of computing and the topics covered provided vital information. I am grateful for the opportunity to learn skills that are applicable within the technology industry and look forward to the opportunity to put them to use.

Reference

Jill West. 2022, 2019. *CompTIA Network+. 9th Edition*.
Retrieved from Cengage Learning, Inc.

TP-Link Website. 2024.
https://www.tp-link.com/us/support/emulators

Router Passwords Website. 2021. By
https://www.routerpasswords.com/

Damon Garn. 15 Oct 2021. Tech Target Website.
https://www.techtarget.com/searchnetworking/tip/Static-IP-vs-dynamic-IP-addresses-Whats-the-difference